

# **Mathematics Bootcamp**

## **Introduction to Proofs and Choice Theory**

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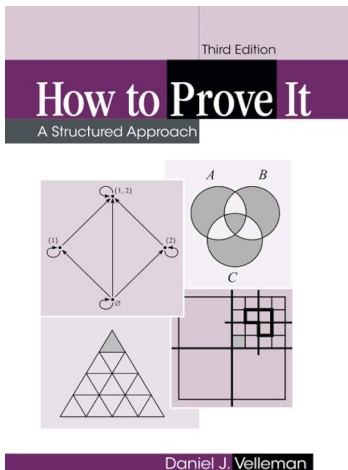
# Logic and Precision

- From Ordinary Language to Scientific Language.
- Symbols, Ideas and the Real world.
- Ideas, Truth Value and Obscurantism.

## These Slides

- In general, I will use the slides to save time and keep notation for the theorems: straight book screenshots.
- But most of the work will be in the board.  
**We learn a language by practice:** trial and error.
- Some believe that studying measure theory is a great way to start. This is like syntactic analysis for a baby in his first words.

# Just Prove It.



- Mathematical statements have a **truth value**: each is either true or false.
- Logic tells us which conclusions follow from stated assumptions.
- Most of what follows is taken from Velleman's book.

# Structured Programming

do

if [condition]

[List of instructions goes here.]

else

[Alternative list of instructions goes here.]

while [condition]

# Proofs as Structured Programming

Let  $x$  be arbitrary.

Suppose  $P(x)$  is true.

[Proof of  $Q(x)$  goes here.]

Thus, if  $P(x)$  then  $Q(x)$ .

Thus, for all  $x$ , if  $P(x)$  then  $Q(x)$ .

# Logical Connectives

**Conjunction:**  $P \wedge Q$

| $P$ | $Q$ | $P \wedge Q$ |
|-----|-----|--------------|
| $T$ | $T$ | $T$          |
| $T$ | $F$ | $F$          |
| $F$ | $T$ | $F$          |
| $F$ | $F$ | $F$          |

**Disjunction:**  $P \vee Q$

| $P$ | $Q$ | $P \vee Q$ |
|-----|-----|------------|
| $T$ | $T$ | $T$        |
| $T$ | $F$ | $T$        |
| $F$ | $T$ | $T$        |
| $F$ | $F$ | $F$        |

**Negation:**  $\neg P$

| $P$ | $\neg P$ |
|-----|----------|
| $T$ | $F$      |
| $F$ | $T$      |

# Quantifiers and Negation

## Quantifiers

Fix a domain  $X$ . A quantifier binds the variable  $x$ :

$\forall x \in X P(x) :$   $P(x)$  holds for every  $x \in X$ ,

$\exists x \in X P(x) :$   $P(x)$  holds for at least one  $x \in X$ .

An occurrence of  $x$  outside the scope of a quantifier is *free*.

**Negation laws** ( $\iff$  means “logically equivalent”)

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q,$$

$$\neg(P \vee Q) \iff \neg P \wedge \neg Q,$$

$$\neg[\exists x \in X P(x)] \iff \forall x \in X \neg P(x), \quad \neg[\forall x \in X P(x)] \iff \exists x \in X \neg P(x).$$

## Implication: $P \rightarrow Q$

| $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-------------------|
| $T$ | $T$ | $T$               |
| $T$ | $F$ | $F$               |
| $F$ | $T$ | $T$               |
| $F$ | $F$ | $T$               |

$P \rightarrow Q$  is false only when  $P$  is true and  $Q$  is false.

## The Contrapositive Is Equivalent

| $P$ | $Q$ | $\neg Q$ | $\neg P$ | $\neg Q \rightarrow \neg P$ | $P \rightarrow Q$ |
|-----|-----|----------|----------|-----------------------------|-------------------|
| $T$ | $T$ | $F$      | $F$      | $T$                         | $T$               |
| $T$ | $F$ | $T$      | $F$      | $F$                         | $F$               |
| $F$ | $T$ | $F$      | $T$      | $T$                         | $T$               |
| $F$ | $F$ | $T$      | $T$      | $T$                         | $T$               |

$P \rightarrow Q$  and  $\neg Q \rightarrow \neg P$  have identical truth values. Thus either statement may be proved by proving the other.

## Proof by Contradiction

| $P$ | $Q$ | $\neg Q$ | $P \wedge \neg Q$ | $\neg(P \wedge \neg Q)$ | $P \rightarrow Q$ |
|-----|-----|----------|-------------------|-------------------------|-------------------|
| $T$ | $T$ | $F$      | $F$               | $T$                     | $T$               |
| $T$ | $F$ | $T$      | $T$               | $F$                     | $F$               |
| $F$ | $T$ | $F$      | $F$               | $T$                     | $T$               |
| $F$ | $F$ | $T$      | $F$               | $T$                     | $T$               |

### Method for proving $P \rightarrow Q$

Assume  $P$  and  $\neg Q$ . If these assumptions imply a contradiction  $\perp$ , then  $P \wedge \neg Q$  is impossible. Therefore  $P \rightarrow Q$ .

# Mathematical Induction

## Principle of Mathematical Induction

Let  $\mathbb{N} = \{1, 2, 3, \dots\}$  and fix  $n_0 \in \mathbb{N}$ . For each natural number  $n \geq n_0$ , let  $P(n)$  be a statement. Suppose that:

1.  $P(n_0)$  is true.
2. For every  $k \geq n_0$ , if  $P(k)$  is true, then  $P(k + 1)$  is true.

Then  $P(n)$  is true for every natural number  $n \geq n_0$ .

# From Logic to Set Theory

## Definition 1.4.1

The *intersection* of two sets  $A$  and  $B$  is the set  $A \cap B$  defined as follows:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

The *union* of  $A$  and  $B$  is the set  $A \cup B$  defined as follows:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

The *difference* of  $A$  and  $B$  is the set  $A \setminus B$  defined as follows:

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

# Set Theory

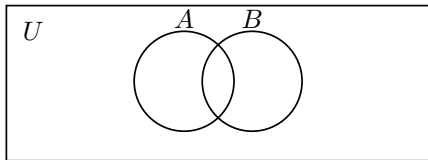


Figure 1.7.

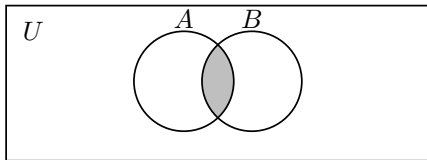


Figure 1.8.  $A \cap B$ .

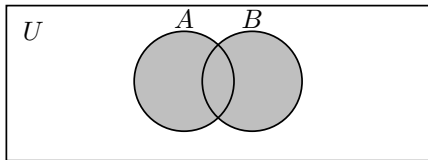


Figure 1.9.  $A \cup B$ .

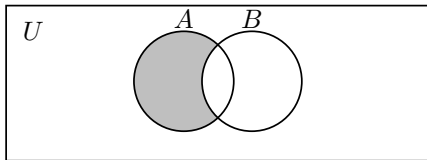


Figure 1.10.  $A \setminus B$ .

# Set Theory

## Definition 1.4.5

Suppose  $A$  and  $B$  are sets. We will say that  $A$  is a *subset* of  $B$  if every element of  $A$  is also an element of  $B$ . We write  $A \subseteq B$  to mean that  $A$  is a subset of  $B$ .  $A$  and  $B$  are said to be *disjoint* if they have no elements in common. Note that this is the same as saying that the set of elements they have in common is the empty set, or in other words  $A \cap B = \emptyset$ .

- Given  $A \subseteq U$ , its *complement in  $U$*  is

$$A^c = U \setminus A = \{x \in U \mid x \notin A\}.$$

- De Morgan's Laws also apply in Sets:

$$(A \cup B)^c = A^c \cap B^c,$$

$$(A \cap B)^c = A^c \cup B^c.$$

# Set Theory

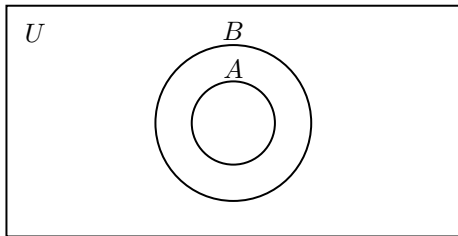


Figure 1.14.  $A \subseteq B$ .

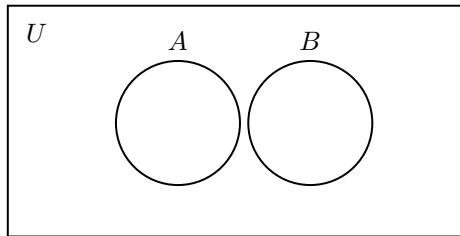


Figure 1.15.  $A \cap B = \emptyset$ .

# Set Theory

## Definition 4.1.1

Suppose  $A$  and  $B$  are sets. Then the *Cartesian product* of  $A$  and  $B$ , denoted  $A \times B$ , is the set of all ordered pairs in which the first coordinate is an element of  $A$  and the second is an element of  $B$ . In other words,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

# Binary Relations

## Definition 4.2.1

Suppose  $A$  and  $B$  are sets. A set  $R \subseteq A \times B$  is a *relation from  $A$  to  $B$* . A *relation on  $A$*  is a set  $R \subseteq A \times A$ .

## Properties of a Relation on $A$

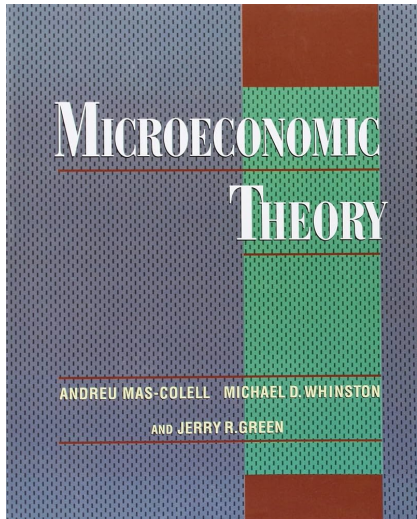
For a relation  $R$  on  $A$ :

1. *Reflexive*:  $xRx$  for every  $x \in A$ .
2. *Complete*:  $xRy$  or  $yRx$  (or both) for every  $x, y \in A$ .
3. *Symmetric*:  $xRy \rightarrow yRx$  for every  $x, y \in A$ .
4. *Transitive*:  $(xRy \wedge yRz) \rightarrow xRz$  for every  $x, y, z \in A$ .

# Time for Economics

- We have seen some mathematical objects and their properties.
- More importantly, these are the words in the formal language that we use in Economics.
- Let's try to read some Economics.

# MWG: Microeconomic Theory



- Direct Screenshots:  
*Feel* that you can learn from it.
- This should not be your unique resource.
- In my 1st year, I also used:
  - **Levin Teaching Notes**
  - Rubinstein Notes Book
  - Kreps Books: 1990, 2012

## Preference and Choice

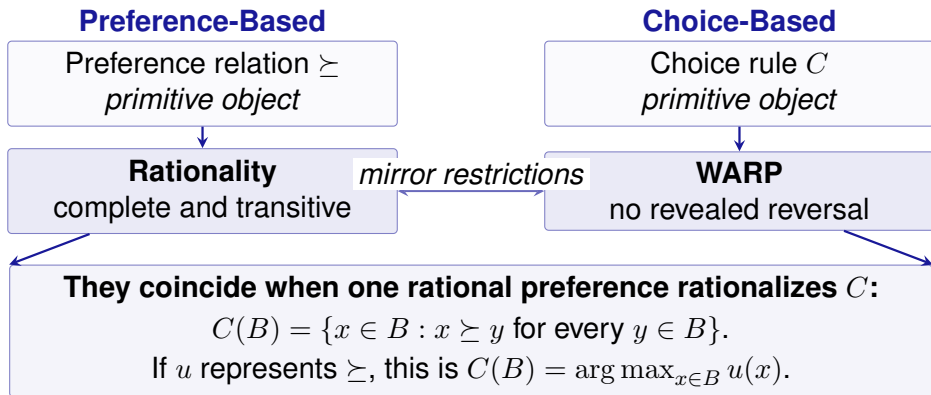
### 1.A Introduction

In this chapter, we begin our study of the theory of individual decision making by considering it in a completely abstract setting. The remaining chapters in Part I develop the analysis in the context of explicitly economic decisions.

The starting point for any individual decision problem is a *set of possible (mutually exclusive) alternatives* from which the individual must choose. In the discussion that follows, we denote this set of alternatives abstractly by  $X$ .

# The Two Approaches Meet at Rationalization

Each approach begins with a different primitive and imposes its own consistency restriction.



# Preference Is the Primitive Object

## A Preference Relation Is the Starting Point

A preference relation  $\succeq$  on  $X$  records the decision maker's ranking:

$$x \succeq y \iff x \text{ is at least as good as } y.$$

- The ranking is taken as given rather than inferred from observed menus.
- **Rationality** restricts the primitive relation: it must be complete and transitive.
- A **utility function** is a numerical representation of that same ranking, when such a representation exists.

## Preference-Based Logic

Start with preferences, impose rationality, and derive choice behavior.

# Preference-Based Approach

## 1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by  $\succsim$ . Technically,  $\succsim$  is a binary relation on the set of alternatives  $X$ , allowing the comparison of pairs of alternatives  $x, y \in X$ . We read  $x \succsim y$  as “ $x$  is at least as good as  $y$ .” From  $\succsim$ , we can derive two other important relations on  $X$ :

- (i) The *strict preference* relation,  $\succ$ , defined by

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

and read “ $x$  is preferred to  $y$ .”<sup>1</sup>

- (ii) The *indifference* relation,  $\sim$ , defined by

$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

and read “ $x$  is indifferent to  $y$ .”

## Preference-Based Approach

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation  $\succsim$ : *completeness* and *transitivity*.<sup>2</sup>

**Definition 1.B.1:** The preference relation  $\succsim$  is *rational* if it possesses the following two properties:

- (i) *Completeness*: for all  $x, y \in X$ , we have that  $x \succsim y$  or  $y \succsim x$  (or both).
- (ii) *Transitivity*: For all  $x, y, z \in X$ , if  $x \succsim y$  and  $y \succsim z$ , then  $x \succsim z$ .

- *Do not waste any attention* speculating about rationality (yet!).
- **Assume** a DM with rational preferences and see where it leads...

**This is exactly the approach we follow.**

# Preference-Based Approach

**Proposition 1.B.1:** If  $\succsim$  is rational then:

- (i)  $\succ$  is both *irreflexive* ( $x \succ x$  never holds) and *transitive* (if  $x \succ y$  and  $y \succ z$ , then  $x \succ z$ ).
- (ii)  $\sim$  is *reflexive* ( $x \sim x$  for all  $x$ ), *transitive* (if  $x \sim y$  and  $y \sim z$ , then  $x \sim z$ ), and *symmetric* (if  $x \sim y$ , then  $y \sim x$ ).
- (iii) if  $x \succ y \succsim z$ , then  $x \succ z$ .

- We can prove this proposition. Let's do it.

# Choice Behavior Is the Primitive Object

## A Choice Rule Is the Starting Point

For each observed menu  $B$ , the rule records a nonempty choice set:

$$C : \mathcal{B} \rightarrow 2^X \setminus \{\emptyset\}, \quad \emptyset \neq C(B) \subseteq B.$$

- Choices are taken as given rather than derived from an unobserved ranking.
- **WARP** restricts the primitive rule by ruling out revealed reversals.
- Revealed preference is constructed from the observed choice behavior.

## Choice-Based Logic

*Do not waste any attention* speculating about rationality (yet!): first record choices, then ask whether one rational relation explains them.

# Choice-Based Approach

## 1.C Choice Rules

In the second approach to the theory of decision making, choice behavior itself is taken to be the primitive object of the theory. Formally, choice behavior is represented by means of a *choice structure*. A choice structure  $(\mathcal{B}, C(\cdot))$  consists of two ingredients:

(i)  $\mathcal{B}$  is a family (a set) of nonempty subsets of  $X$ ; that is, every element of  $\mathcal{B}$  is a set  $B \subset X$ . By analogy with the consumer theory to be developed in Chapters 2 and 3, we call the elements  $B \in \mathcal{B}$  *budget sets*.

(ii)  $C(\cdot)$  is a *choice rule* (technically, it is a correspondence) that assigns a nonempty set of chosen elements  $C(B) \subset B$  for every budget set  $B \in \mathcal{B}$ . When  $C(B)$  contains a single element, that element is the individual's choice from among the alternatives in  $B$ . The set  $C(B)$  may, however, contain more than one element. When it does, the elements of  $C(B)$  are the alternatives in  $B$  that the decision maker *might* choose; that is, they are her *acceptable alternatives* in  $B$ . In this case, the set  $C(B)$

# What Is a Rich Domain of Menus?

## Every Small Menu Is Observable

The domain is *rich* if every nonempty  $B \subseteq X$  with  $|B| \leq 3$  belongs to  $\mathcal{B}$ . Thus, for distinct  $x, y, z \in X$ , it contains

$$\{x\}, \{y\}, \{z\}; \quad \{x, y\}, \{x, z\}, \{y, z\}; \quad \{x, y, z\}.$$

- **Singletons** anchor choice:  $C(\{x\}) = \{x\}$ .
- **Pairs** recover complete pairwise comparisons.
- **Triples** bring three comparisons together and test transitivity.

## Why “At Most Three”?

Pairs build comparisons; triples test them. Larger menus are allowed.

# Why the Proof Starts with Pairs

## Pairs Recover Revealed Preference

Suppose every pair belongs to  $\mathcal{B}$  and choice is nonempty. Under WARP,

$$x \succeq^* y \iff x \in C(\{x, y\}).$$

- $\Leftarrow$  is immediate: the pair itself is an observed menu.
- For  $\Rightarrow$ ,  $x$  was chosen somewhere while  $y$  was available. If the pair chooses  $y$ , WARP forces it to choose  $x$  as well.
- Since every pair has a choice, either  $x \succeq^* y$  or  $y \succeq^* x$ . Hence  $\succeq^*$  is complete.

# Choice-Based Approach

**Definition 1.C.1:** The choice structure  $(\mathcal{B}, C(\cdot))$  satisfies the *weak axiom of revealed preference* if the following property holds:

If for some  $B \in \mathcal{B}$  with  $x, y \in B$  we have  $x \in C(B)$ , then for any  $B' \in \mathcal{B}$  with  $x, y \in B'$  and  $y \in C(B')$ , we must also have  $x \in C(B')$ .

**Definition 1.C.2:** Given a choice structure  $(\mathcal{B}, C(\cdot))$  the *revealed preference relation*  $\succsim^*$  is defined by

$x \succsim^* y \Leftrightarrow$  there is some  $B \in \mathcal{B}$  such that  $x, y \in B$  and  $x \in C(B)$ .

# Why the Proof Needs Triples

## A Cycle Hidden in Pair Menus

$$C(\{x, y\}) = \{x\}, \quad C(\{y, z\}) = \{y\}, \quad C(\{x, z\}) = \{z\}.$$

With pairs alone, WARP does not connect these observations: no two distinct pair menus contain the same two alternatives. Yet the choices form the cycle

$$x \text{ over } y, \quad y \text{ over } z, \quad z \text{ over } x.$$

## Add the Triple $T = \{x, y, z\}$

If  $T$  chooses  $x$ ,  $y$ , or  $z$ , that alternative loses in one of the pair menus. WARP would require it to be chosen from that pair too. Thus a cycle would force  $C(T) = \emptyset$ , contradicting nonempty choice.

# Rationalizing Choice

## Rationalization

A preference relation  $\succeq$  *rationalizes* a choice rule  $C$  on  $\mathcal{B}$  if, for every  $B \in \mathcal{B}$ ,

$$C(B) = \{x \in B : x \succeq y \text{ for every } y \in B\}.$$

The choice rule is *rationalizable* if some rational preference relation rationalizes it.

Thus, observed choice contains exactly the best available alternatives according to one complete and transitive relation.

# How the Rationalization Proof Works

## WARP and Rationalizability

Suppose  $\mathcal{B}$  contains every nonempty subset of  $X$  with at most three alternatives, and  $C(B) \neq \emptyset$  for every  $B \in \mathcal{B}$ . Then  $C$  is rationalizable if and only if it satisfies WARP.

1. Pair menus recover  $\succeq^*$  and make it complete.
2. Triple menus and WARP rule out cycles, making  $\succeq^*$  transitive.
3. WARP then links pairwise comparisons back to every observed menu:

$$C(B) = \{x \in B : x \succeq^* y \text{ for every } y \in B\}.$$

## The Key Step

If  $y \in C(B)$  and  $x \succeq^* y$ , the pair menu chooses  $x$ ; WARP therefore forces  $x \in C(B)$  too.